Menstrual health among female university students

Project brief | Health ecology, statistical analysis and machine-learning evaluation

Research question and course project

How are environmental, social, behavioural, psychological and individual characteristics associated with menstrual health, and how well can questionnaire-based models distinguish respondents with and without the study outcomes?

In the Field Practice of Epidemiology module, the course team designed a questionnaire under a health ecology framework and analysed **436 valid responses**. The original project combined descriptive statistics, between-group comparisons, multivariable logistic regression, and decision tree, random forest, Naive Bayes and support vector machine classifiers.

Reanalysis and evaluation

The repository builds on that team project by reconstructing outcomes from the questionnaire fields and comparing five classifiers, including logistic regression as a prediction baseline. It uses five-fold stratified cross-validation repeated ten times. Imputation, encoding and top-30 feature selection are fitted on training folds only; model settings are fixed.

Primary outcome: the final course definition: cycle length outside 21–35 days, bleeding duration outside 3–7 days, PBAC below 37 or above 150, or reported dysmenorrhea. Any one qualifies (384 positive / 52 negative).

Secondary outcome: dysmenorrhea alone (269 / 167). **Sensitivity analysis:** cycle or bleeding abnormality, using the first three criteria without dysmenorrhea (288 / 148). These are questionnaire-based operational definitions, not clinical diagnoses.

Additional analyses include a 70-test univariable scan with Benjamini–Hochberg correction within each outcome, exploratory 16-predictor multivariable logistic models, a **permutation test of the logistic-regression pipeline only**, and a dysmenorrhea analysis excluding two family-history questions.

The repository presents the subsequent reanalysis; it does not attribute these additional validation steps to the original course work. No individual survey records are published.

Results and interpretation

Primary composite: mean cross-validated ROC-AUC ranged from 0.46 to 0.58 across five classifiers. The logistic-regression permutation test gave **p = 0.54** (200 shuffles, one five-fold split), providing no evidence that this logistic pipeline outperformed its shuffled-label reference. This p-value does not test the other classifiers.

Dysmenorrhea: mean ROC-AUC ranged from 0.63 to 0.68. After removing the two family-history questions it ranged from 0.55 to 0.56. This describes a decrease in predictive performance, not proof that family history is the only relevant factor or a causal mechanism.

Cycle or bleeding abnormality: evaluated models showed near-chance discrimination in this sample (ROC-AUC 0.49–0.53). No statistically significant association with insecticide use was detected in the univariable analyses of any of the three outcomes (p = 0.18–0.97). These results do not establish that the outcome is inherently unpredictable or that an association is absent.

Associations: maternal dysmenorrhea history was the only variable retained after BH correction in the primary-outcome univariable scan. In a *separate* multivariable model, its covariate-adjusted OR was 7.4 (95% CI 2.6–21.3); for dysmenorrhea alone it was 6.9 (4.0–11.9). Covariate adjustment and multiple-testing correction refer to different analyses.

Limits and learning

Only 52 respondents are negative for the primary composite. Its 16-predictor logistic model may produce unstable estimates. Cross-validation folds overlap, so their standard deviation describes spread and is not a confidence interval. Associations are exploratory; the cross-sectional, self-reported convenience sample cannot establish causation or population prevalence.

The project highlights the importance of outcome definition, data quality, sample distribution and evaluation design when applying machine learning to health data. The models are not validated screening tools. Full tables, methods and limitations are provided in the repository.

Survey and original analyses: course team. This repository: subsequent reanalysis and documentation. Python tools include pandas, scikit-learn, statsmodels and SciPy.